

Abdul Monaf Chowdhury

[Website](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

University of Dhaka, Bangladesh

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EDUCATION

Bachelor of Science in Robotics and Mechatronics Engineering

Jan 2019 — Jan 2024

University of Dhaka, Bangladesh

CGPA: **3.87/4.00**, Ranked 2nd place

Relevant Coursework:

Artificial Intelligence, Introduction to Machine Learning, Digital Image Processing and Robot Vision, Digital Signal Processing, Human-Robot Interaction, Advanced Robotics

Skills:

Software: C, C++, Python, MATLAB, PyTorch, JAX, Flax, TensorFlow, $\text{\LaTeX}$

Language: Fluent in both English and Bangla

Test Scores:

GRE: **318** (Quant 163, Verbal 155, Analytical 4.0)

IELTS: **8.0** (R 9.0, L 8.5, S 6.5, W 7.5)

RESEARCH INTERESTS

Multi-modal Learning, Computer Vision, Embodied AI, Vision Language Models, Reinforcement Learning

RESEARCH EXPERIENCE

Research Assistant

Feb 2024 – Dec 2025

MAIM Lab, University of Dhaka

Dhaka, Bangladesh

Funding: *Wellcome Leap (In Utero, California, USA)*

PI: *Dr. Niamh Nowlan*, Co-PI: *Dr. Abhishek Kumar Ghosh*

- Collaborated with **University College Dublin** on the **Wellcome Leap In Utero** funded project titled “Translation of a Wearable Fetal Movement Monitor towards Stillbirth Prevention”
- Designed and implemented deep learning-based frameworks to analyze multimodal sensor signals from wearable belts to detect body movements, fetal kicks, and fetal hiccups
- Optimized signal processing of sensor data, presented hardware design feedback based on analytical findings, and validated hardware design changes
- Assisted in attaining **\$1M** funding extension, and eventually helped secure translational funding from Wellcome Leap to launch a **startup**

Research Assistant

Jan 2023 – Jan 2024

AVIoT Lab, University of Dhaka

Dhaka, Bangladesh

Supervisor: *Dr. Md Mehedi Hasan*; *[Project Report]*

- Collaborated on Unmanned Aerial Vehicle (UAV) based suspicious human activity recognition and drone surveillance
- Designed a hybrid model combining modified 3D CNN and FFT-based action recognition module for drone surveillance applications
- Built a lightweight deep learning pipeline using MobileNetV2 + BiLSTM for edge-based human activity detection, significantly reducing inference time

PROFESSIONAL EXPERIENCE

Instructor

May 2025 – Aug 2025

National Camp, Bangladesh AI Olympiad

Dhaka, Bangladesh

- Instructed national camp students on Unsupervised Learning, Deep Learning, and Computer Vision algorithms and architectures, took relevant labs, and illustrated Deep Learning evaluation strategies and techniques
- Planned, organized, and executed the AI problem challenge competition on Kaggle to select the final four candidate to represent Bangladesh in the International Olympiad on AI 2025

PUBLICATIONS

- *LaGEA: Language Guided Embodied Agents for Robotic Manipulation* [\[Paper\]](#) [\[Code\]](#)
Abdul Monaf Chowdhury, AKM Moshir Rahman Mazumder, Rabeya Akter Fariya, Safaeid Hossain
43rd International Conference on Machine Learning, ICML '26
- *T3Time: Tri-Modal Time Series Forecasting via Adaptive Multi-Head Alignment and Residual Fusion* [\[Paper\]](#) [\[Code\]](#)
Abdul Monaf Chowdhury, Rabeya Akter Fariya, Safaeid Hossain
40th Annual AAAI Conference on Artificial Intelligence, AAAI '26
- *U-ActionNet: Dual-Pathway Fourier Networks with Region-of-Interest Module for Efficient Action Recognition in UAV Surveillance* [\[Paper\]](#)
Abdul Monaf Chowdhury, Ahsan Imran, Md Mehedi Hasan, Riad Ahmed, AKM Azad, Salem A. Alyami
IEEE Access, 2024. IF - 3.4
- *FFT-UAVNet: FFT Based Human Action Recognition for Drone Surveillance System* [\[Paper\]](#)
Abdul Monaf Chowdhury, Ahsan Imran, Md Mehedi Hasan
5th IEEE International Conference on Sustainable Technologies for Industry 5.0 (STI), 2023

MANUSCRIPT SUBMITTED

- *Counting Through Occlusion: Framework for Open World Amodal Counting* [\[Paper\]](#) [\[Code\]](#)
Safaeid Hossain, Rabeya Akter Fariya, **Abdul Monaf Chowdhury**, Md Mehedi Hasan, Md. Jubair Ahmed Surov
European Conference on Computer Vision - ECCV 2026

RESEARCH PROJECT

Open World Amodal Counting[\[Code\]](#) [\[Paper\]](#)

Aug 2025 – Nov 2025

Multimodal Learning, Vision Language Model, Representation Learning

- Architected a hierarchical Feature Reconstruction Module that fuses visible spatial context with vision–language priors to reconstruct class-discriminative features at occluded locations, enabling amodal counts from RGB images
- Designed a Visual-Equivalence consistency objective with teacher–student alignment, anchoring reconstructions to unoccluded features and restoring robustness without depth sensors or structured layouts
- Built occlusion-augmented FSC-147 and CARPK benchmarks; achieved SOTA results under occlusion where MAE reduced by **32.2%↓** (FSC-147), **40.7%↓** (CARPK), **40.0%↓** (CAPTURE-Real), demonstrating strong cross-dataset generalization

Language Guided Embodied Agents[\[Code\]](#) [\[Paper\]](#)

Mar 2025 – Sep 2025

Embodied AI, Vision Language Model, Reinforcement Learning

- Designed and integrated a **Qwen 2.5VL-3B** VLM driven “episodic reflection” module, automatically generating rich, natural-language self-assessments of each trial, highlighting successes and pinpointing failure causes to provide the agent with human-like introspection
- Fused multimodal reward signals by combining these verbal reflections with CLIP-style vision–language feedback from task descriptions and goal images, crafting a dense, semantically grounded reward model
- Engineered a reward-aligned **Soft Actor Critic**-based learning pipeline, where the enriched feedback loop accelerated exploration and policy refinement, consistently reducing training time and reliably converging on optimal behaviours across the Meta-World manipulation tasks

Tri-Modal Time Series Forecasting [\[Code\]](#) [\[Paper\]](#)

Apr 2025 – Aug 2025

Large Language Model, Deep Learning, Signal Processing

- Architected an Adaptive Dynamic Multi-Head Cross-Modal Attention module with channel-wise residual skip-connections, enabling fine-grained alignment between temporal and auxiliary features and boosting representational capacity across modalities
- Engineered an **FFT**-based Frequency-Domain Processing pipeline, projecting real-valued spectra into learnable tokens and applying transformer-based attention with weighted pooling to extract robust spectral embeddings for each sensor channel
- Designed a Trainable Adaptive Rich-Horizon Gating Fusion to dynamically combine spectral and temporal encodings—replacing naive concatenation—and beat the state-of-the-art benchmark on multivariate time-series forecasting

Proxemics & Social Interaction Patterns in ASD Children

Sep 2023 — Jan 2024

Human-Robot Interaction, Deep Learning

- Formulated a **YOLOv8**-based system to determine the ideal proxemics of autism spectrum disorder (ASD) children in front of **NAO** Robot
- Examined and analyzed the behavioural responses of twenty children diagnosed with ASD in the presence of specific actions performed by the NAO robot

Automatic Stock Trading **[Report]**

Aug 2022 — Nov 2022

Reinforcement Learning

- Implemented Approximate Q Learning for three Bangladeshi stocks to generate Buy, Sell, and Hold orders
- Achieved 11% return of investment for the 3 stocks beating the DSE 30 index

AWARDS & SCHOLARSHIPS

- Dean's Award - for best Undergraduate Result, University of Dhaka, 2024
- Engineering Faculty Undergraduate Merit Scholarship, University of Dhaka, 2024
- 5th, Dataverse Challenge - ITVerse, Bangladesh, 2023; [\[Report\]](#)
- 2nd, Intra-Department Soccer Bot Championship, University of Dhaka, 2019
- Sylhet Board Scholarship, Higher Secondary Certificate Examination 2018

WORKSHOP/CONFERENCE ATTENDED

- 5th International Conference on Sustainable Technologies for Industry 5.0 (STI), Dhaka
- 40th Annual AAAI Conference on Artificial Intelligence, AAAI '26

REVIEWER

- IEEE Access
- AAAI 26

LEADERSHIP/VOLUNTEER ACTIVITIES

General Secretary

Mar 2022 — Feb 2024

RMEDU Student Club, University of Dhaka

Academic Team Mentor

Sep 2019 — Aug 2022

Bangladesh Robot Olympiad

Program Co-Ordinator

Jul 2021 -- Jun 2022

IEEE Robotics & Automation Society, University of Dhaka